Heart-brain axis: Exercise Mitigates Oxidative Stress–Driven Cardiac and Neurovascular Dysfunction And It May Enhance Amyloid-β Clearance in Alzheimer's Disease Mice.

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Alzheimer Disease (AD) is increasingly viewed as a systemic disorder in which cardiometabolic dysfunction and neurodegeneration are linked through chronic inflammation and oxidative stress within a heart-brain axis (HBA). Cardiometabolic risk factors such as diabetes, obesity, and hypertension drive overproduction of reactive oxygen species (ROS), endothelial dysfunction, and systemic inflammation, which impair blood-brain barrier (BBB) integrity and exacerbate amyloid-β (Aβ) pathology. This study proposes to investigate whether moderate treadmill exercise reduces oxidative stress-driven cardiac and neurovascular dysfunction and, thereby, enhances Aβ clearance in 5xFAD mice. Male transgenic and wild-type mice will undergo sedentary or exercise conditions for 8-12 weeks beginning at 4 months of age. Cardiac function will be assessed longitudinally by transthoracic echocardiography. Oxidative stress will be quantified in heart tissue using (DHE) fluorescence and MDA assays. Brain, heart and blood tissues will be harvested from paired animals to evaluate ROS levels, cytokine expression, neuroinflammation and BBB integrity. We hypothesized that exercise will improve cardiac performance, reduce oxidative stress, preserve BBB tight junctions, attenuate glial activation, and promote Aβ clearance. Two-way ANOVA will test genotype and exercise effect ($p < 0.05$). This work aims to clarify how exercise-induced modulation of the HBA may counteract oxidative and inflammatory mechanisms that couple cardiometabolic dysfunction to AD progression.

Keywords: Alzheimer, Heart-Brain Axis, Cardiovascular Dysfunction, Inflammation, Blood–Brain Barrier (BBB), Reactive Oxygen Species (ROS), Physical Exercise

Introduction

Neurodegenerative diseases such as Alzheimer's disease (AD) and Parkinson's disease (PD) are leading causes of disability and death in aging populations; current therapies are largely symptomatic and don't stop the progression. A unifying feature across these disorders is chronic neuroinflammation, characterized by the accumulation of misfolded proteins like amyloid-β, tau, and α-synuclein [1–3]. At the interface between body and brain, the blood–brain barrier (BBB) plays a central role. However, during peripheral inflammation, its function becomes impaired as multiple inflammatory pathways are activated, leading to structural and functional alterations that allow cytokines and immune cells into the central nervous system [4,5]. Increasing evidence suggests that neuroinflammation is closely linked to peripheral physiological disturbances, particularly cardiovascular dysfunction and systemic low-grade inflammation [6,7]. Cardiometabolic risk factors- diabetes, obesity, and hypertension are consistently associated with overproduction of reactive oxygen and nitrogen species (ROS/RNS) during oxidative metabolism that lead to elevations of pro-inflammatory cytokines such as TNF-α, IL-6, and IL-1β [6–8]. Chronic inflammation also directly impacts protein homeostasis. Pro-inflammatory signaling through pathways such as NF-κB and the NLRP3 inflammasome enhances oxidative stress, alters post-translational modifications of tau, and promotes aggregation and prion-like propagation of misfolded proteins [3,8].

By causing decreased blood and a change in the neurovascular apparatus, heart disease, especially heart failure lead to the development and progression of neurodegenerative diseases like AD [9]. Besides damaging the blood-brain barrier and triggering inflammation in the central nervous system, oxidative stress profoundly affects the heart by causing changes in

cellular macromolecules and subsequently, inducing modification at subcellular organelles, like sarcolemma, sarcoplasmic reticulum, nucleus, and mitochondria. During oxidative stress, reactive oxygen species (ROS) contribute to endothelial dysfunction and vascular impairment by interfering with the protective nitric oxide (NO) signaling pathway, ultimately resulting in nitric oxide synthase (NOS) uncoupling [9]. Atherosclerosis and AD share overlapping inflammatory mechanisms, including complement activation at sites of endothelial injury and in response to aggregated proteins, supporting the concept of a systemic inflammatory milieu that simultaneously damages the cardiovascular system and the brain [7,12]. It has been reported that approximately one-third of dementia cases linked to AD are associated with cardiovascular events. Cerebral hypoperfusion and vascular degeneration play a role in various stages of AD. Consequently, disruptions in the blood-brain barrier (BBB) represent an early pathology that can hinder the clearance of Aβ and favours its transfer from the periphery into CNS [10].

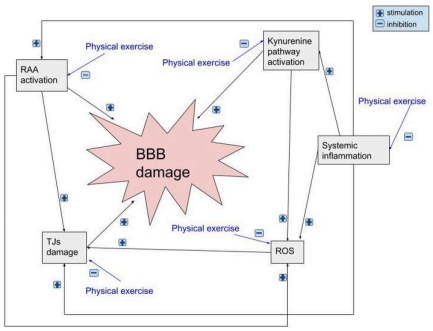

Figure 1. Low-grade systemic inflammation contributes to BBB damage via pro-inflammatory cytokines leading to neurotoxic processes on BBB and its permeability [28].

Fortunately, these processes are modifiable. Physical exercise and other lifestyle interventions reduce cardiometabolic risk, improve vascular function, and lower systemic levels of pro-inflammatory cytokines, while enhancing antioxidant defenses and neurotrophic signaling [11,12]. Exercise can attenuate microglial overactivation, improve BBB integrity, and promote proteostatic mechanisms that limit pathogenic protein aggregation, suggesting a plausible route to protect cardiovascular and brain health simultaneously.

Cardiovascular and metabolic disturbances may act as upstream drivers of a shared cytokine-mediated cascade that connects body and brain in neurodegeneration, and this area is not well explored. In this study we aimed to investigate the link between cardiometabolic risk factors, associated overproduction of ROS, and their role in developing heart and neurodegenerative diseases.

Previous studies show that regular physical exercise exerts pleiotropic neuroprotective effects in AD models and patients, improving cognition, reducing amyloid burden and tau pathology, and attenuating oxidative stress and neuroinflammation, largely via modulation of mitochondrial function, antioxidant pathways, and microglial activation [13].

Hypothesis

Exercise reduces pathological ROS accumulation disrupting the expression of pro-inflammatory cytokines TNF-α, IL-1β, IL-6, which improves cardiac function in Alzheimer's disease mice. Ultimately, this leads to lowering systemic inflammation, preserving blood-brain barrier integrity and attenuating neuroinflammation by improving the clearance of amyloid-β protein from the brain.

Methods

Animal Model and Experimental Design

Transgenic 5xFAD mice will be used as a model of Alzheimer's disease (AD). These mice carry the Swedish APP mutation and two PSEN1 mutations, leading to accelerated amyloid-β plaque accumulation. Thus, mimicking the key features of AD in humans, including neuroinflammation, synapse disruption and cognitive decline. Disease features begin to emerge around 4 months of age, allowing assessment of interventions during early pathology development [14]. Thus, the experiment will begin at 4 months old.

Mice will be randomly assigned to four groups: 5xFAD exercise, 5xFAD sedentary, Wild-type (WT) exercise, and Wild-type (WT) sedentary. Only male mice will be included to reduce variability introduced by estrous-cycle hormone fluctuations, which are known to modulate immune signaling, microglial activation, and inflammatory responses in the brain [15]. While this improves statistical power and reproducibility, it represents a limitation of the study, and future work should evaluate sex-specific responses.

Animals will be housed under standard SPF conditions with controlled light cycles, sterilized pellet diet, and water, in compliance with ARRIVE guidelines and IACUC-approved protocols [16, 17].

Exercise Protocol

Exercise groups will undergo moderate-intensity treadmill training on flat treadmills, with a velocity of

12-15 m/min and ~60-75% VO_2max. Training will begin at 4 months of age and continue for 8-12 weeks.

The first week will serve as an acclimation period with gradual speed increases. The following weeks will be a standard protocol consisting of a 5-minute warm-up, 50-minute running phase, and 5-minute cool-down, performed 5 days a week. Sedentary control mice will be placed on stationary treadmills for an equivalent duration without induced running. Body weight and fatigue scores will be monitored weekly to ensure animal welfare and training consistency.

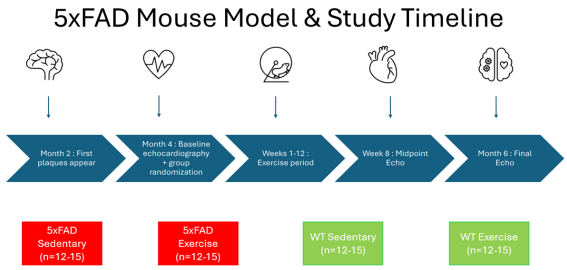

Figure 2. Experiment timeline

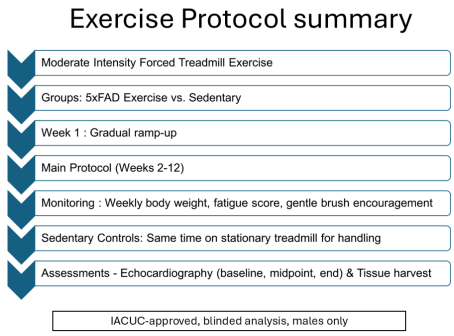

Figure 3. Exercise protocol summary

Cardiac Function Assessment

Cardiac function will be evaluated using transthoracic echocardiography with a high-resolution ultrasound system under light isoflurane anesthesia (~1.5%) [18]. Measurements will be taken in a blinded manner at: baseline (at 4 months), midpoint (after 8 weeks) and endpoint (after 12 weeks). Parameters assessed will include cardiac output, stroke volume, fractional shortening, ejection fraction, and systolic and diastolic function.

Tissue collection

Mice will be processed in paired subcohorts within each group to preserve tissue quality for different analysis. This ensures optimal tissue preservation while maintaining comparable experimental conditions.

One subgroup will undergo terminal cardiac puncture to obtain plasma for cytokine ELISA, followed by transcardial perfusion with PBS and PFA for fixation [19]. Brains from these animals will be used for Thioflavin-S staining and immunohistochemistry.

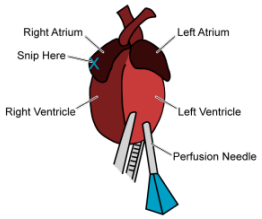

Figure 4. Transcardial perfusion puncture [19]

The paired subgroup will undergo rapid dissection [20,21]. Brain, cortex and hippocampus, and heart tissue will be snap-frozen for biochemical assays including ELISA quantification and assessment of cardiac oxidative stress markers.

Molecular Assessment of ROS, Cytokines, Neuroinflammation, BBB Integrity and Amyloid Clearance

Cardiac oxidative stress will be quantified using dihydroethidium (DHE) fluorescence imaging of unfixed heart tissue with normalized intensity analysis [22]. For the assessment of cytokines, plasma levels of TNF-α, IL-1β, and IL-6 will be quantified using mouse-specific ELISA kits following manufacturer protocols [23].

Neuroinflammation will be assessed using immunostaining for Iba1, to quantify microglial activation, and GFAP, to assess reactive astrocytes. These markers are sensitive to exercise-induced changes in glial activity [24]. Additionally, BBB integrity will be evaluated using Claudin-5 immunostaining, a tight-junction protein essential for BBB structure [25].

Fibrillar amyloid plaques will be quantified using Thioflavin-S staining of fixed brain sections [26]. Soluble and insoluble Aβ40/42 levels will be measured from fresh cortical and hippocampal homogenates using sequential extraction followed by ELISA [27].

Statistical analysis

Data will be analyzed using two-way ANOVA with genotype and exercise as factors, followed by post-hoc comparisons when appropriate.

Two-way ANOVA is routinely used in AD mouse studies examining: amyloid ELISA quantification and plaque burden from histological staining. This statistical analysis allows evaluation of both genotype effects and intervention effects simultaneously.

Hypothesized Results

We expect moderate treadmill exercise to improve cardiac performance in 5xFAD mice, shown by increased ejection fraction, stroke volume, and fractional shortening relative to sedentary AD mice [29]. This overall improvement is expected to coincide with reduced cardiac oxidative stress by lower DHE fluorescence. In previous literature, exercise has been shown to enhance antioxidant defense and suppress ROS accumulation [30], along with decreased systemic inflammation by having lower TNF-α, IL-1β, and IL-6 levels [31,35]. To determine whether exercise significantly alters systemic inflammation, cytokine concentrations could be analyzed using two-way ANOVA in future research.

In the brain, the group of 5xFAD mice who perform exercise are expected to show preserved blood-brain barrier integrity through maintained claudin-5 expression, as physical activity has been linked to improved cerebrovascular stability and tight-junction preservation [28]. This group should also exhibit attenuated neuroinflammation, evidenced by reduced microglial and astrocytic activation measured through Iba1 and GFAP immunostaining. Decreases in Iba1-positive microglia and GFAP-positive astrocytes align with prior findings [32].

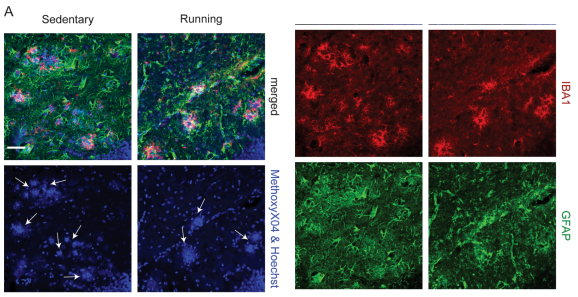

Figure 5. Running decreased hippocampal amyloid burden. Representative confocal images of tissue sections from the hippocampus of sedentary (left) and running (right) APP/PS1 mice, with labeling of astrocytes (anti-GFAP, green), microglia (anti-Iba1, red), amyloid plaques (Methoxy-X04, blue), and cell nuclei (Hoechst, blue). The white arrows in the second column of images indicate Methoxy-X04 labeled amyloid plaques, which are distinguished from Hoechst-labeled cell nuclei by morphological differences. [32].

Additionally, exercise is hypothesized to enhance amyloid-beta clearance by lowering soluble and insoluble Aβ40/42 levels by ELISA [33]. As well as, the clearance should be reflected by reduced fibrillar plaque burden detected with Thioflavin-S staining [34].

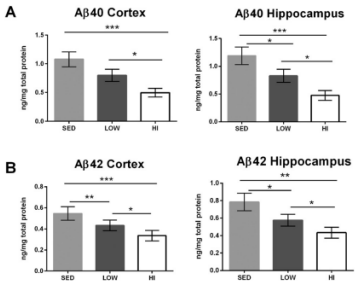

Figure 6. Soluble Aβ concentrations are decreased by exercise training in a dose-dependent manner. (A) Aβx − 40 concentrations in cortex and hippocampus. (B) Aβx − 42 concentrations in cortex and hippocampus. Values represent mean ± s.e.m. *P b 0.05; **P b 0.01; ***P b 0.001. [33]

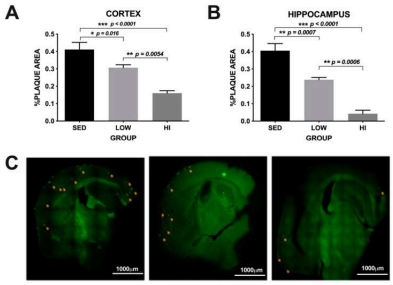

Figure 7. Amyloid plaque % area is decreased in an exercise training dose-dependent manner in both the cortex (Panel A) and hippocampus (Panel B) of Tg2576 mice following 12 months of treadmill running. Panel C shows a representative image of amyloid plaque detected by Thioflavin S staining at 10× magnification [34].

Together, these outcomes would support a HBA mechanism in which exercise reduces systemic inflammation and oxidative stress, thereby promoting neuroprotection and slowing Alzheimer's pathology.

Discussion

The idea that Alzheimer's disease (AD) is a systemic cardiometabolic condition where neurodegeneration is actively driven by oxidative stress and inflammation within the HBA is advanced by this study. Exercise may protect the brain indirectly by blocking upstream peripheral mechanisms that exacerbate neurovascular disease, as seen by the expected improvement in cardiac performance together with decreased myocardial ROS and systemic cytokines. Our findings establish the heart as a modifiable regulator of AD progression rather than a secondary bystander by associating enhanced myocardial function with lower inflammatory signals and preserved brain perfusion.

Exercise stabilizes the neurovascular unit through systemic redox and inflammatory regulation, which is further supported by the preservation of the blood-brain

barrier integrity and the attenuation of glial activation. Improved clearance capacity is implied by a lower soluble and fibrillar Aβ burden, which may be brought about by better vascular function and less inflammatory blockage of proteostatic pathways. The heart-brain axis theory is strengthened by these combined cardiac and neurovascular results, which also imply that addressing cardiometabolic dysfunction rather than focusing solely on central nervous system management may reduce amyloid development through multi-system control.

The amyloid-centric 5xFAD model, the use of only male mice and the lack of behavioural evaluation, which limits direct cognitive interpretation, are some of the limitations. To ascertain if cardiometabolic modulation results in long lasting neuroprotection, future research should include both sexes, long term cognitive testing and more thorough mechanistic analysis of vascular clearance routes. However, this research offers a strong basis for redefining exercise as a systemic treatment approach that can alter the interrelated approach to cardiac and neurovascular causes of Alzheimer's disease.

Conclusion

We believe that this study will provide critical insights into the heart brain interactions that drive Alzheimer's disease progression. Using the 5xFAD mouse model and a structured treadmill intervention, this integrative approach examines how cardiac dysfunction, oxidative stress, inflammation, BBB integrity, and amyloid-β pathology are interconnected. By linking cardiac and neurovascular outcomes, the study will assess whether moderate exercise improves cardiac performance, reduces oxidative and inflammatory damage, enhances Aβ clearance, and protects the neurovascular unit. Overall, the findings may support exercise as a systemic, cardiometabolic strategy to delay AD progression.

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